

$$\begin{aligned}
& \rightarrow \frac{1}{4} (g_1^2 + g_2^2) + \\
& \frac{1}{16\pi^2} \left(\frac{1}{24} (-3g_1^4 - 5g_2^4) + \frac{4}{9} C_{\phi 1^2} g_2^4 LF_{3,0}[m_2] + \frac{2}{3} C_{\phi 1^2} g_2^4 LF_{4,-1}[m_2] - \frac{32}{45} C_{\phi 1^2} g_2^4 LF_{5,-2}[m_2] - \right. \\
& \frac{1}{12} \sum_p g_1^4 LF_{2,0}[m_d^p] - \frac{1}{4} \sum_p g_1^4 LF_{2,0}[m_e^p] - \frac{1}{8} \sum_p (g_1^4 + g_2^4) LF_{2,0}[m_l^p] - \\
& \frac{5}{18} \sum_p C_{\phi 1^2} g_2^4 LF_{3,0}[m_l^p] + \frac{1}{12} \sum_p C_{\phi 1^2} g_2^4 LF_{4,-1}[m_l^p] + \frac{8}{45} \sum_p C_{\phi 1^2} g_2^4 LF_{5,-2}[m_l^p] + \\
& \left. \left(-\frac{1}{2} \overline{y_u^{pr}} y_u^{pr} (g_1^2 - 3g_2^2) - \frac{1}{24} \sum_p (g_1^4 + 9g_2^4) \right) LF_{2,0}[m_q^p] + \right. \\
& \frac{1}{6} C_{\phi 1^2} g_2^2 (36 \overline{y_u^{pr}} y_u^{pr} - 5 \sum_p g_2^2) LF_{3,0}[m_q^p] + \frac{1}{4} C_{\phi 1^2} g_2^2 (-24 \overline{y_u^{pr}} y_u^{pr} + \sum_p g_2^2) LF_{4,-1}[m_q^p] + \\
& \frac{8}{15} \sum_p C_{\phi 1^2} g_2^4 LF_{5,-2}[m_q^p] - \frac{1}{3} \sum_p g_1^4 LF_{2,0}[m_u^p] + 2g_1^2 \overline{y_u^{pr}} y_u^{pr} LF_{2,0}[m_u^r] + \\
& \frac{2}{9} C_{\phi 1^2} g_2^4 LF_{3,0}[\tilde{\mu}] + \frac{1}{3} C_{\phi 1^2} g_2^4 LF_{4,-1}[\tilde{\mu}] - \frac{16}{45} C_{\phi 1^2} g_2^4 LF_{5,-2}[\tilde{\mu}] + \frac{1}{2} g_1^4 LF_{2,2,-2}[m_1, \tilde{\mu}] + \\
& \frac{1}{2} g_1^4 m_1^2 LF_{2,2,-1}[m_1, \tilde{\mu}] - \frac{8}{3} C_{\phi 1^2} g_2^4 LF_{2,1,0}[m_2, \tilde{\mu}] + \frac{5}{2} g_2^4 LF_{2,2,-2}[m_2, \tilde{\mu}] + \\
& \frac{1}{6} g_2^4 (-34 C_{\phi 1^2} + 3m_2^2) LF_{2,2,-1}[m_2, \tilde{\mu}] + \frac{2}{3} m_2 \tilde{\mu} g_2^4 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{2,2,0}[m_2, \tilde{\mu}] + \\
& \frac{4}{3} C_{\phi 1^2} g_2^4 LF_{3,1,-1}[m_2, \tilde{\mu}] + \frac{28}{3} C_{\phi 1^2} g_2^4 LF_{3,2,-2}[m_2, \tilde{\mu}] - \\
& \frac{1}{3} m_2 g_2^4 (18 m_2 C_{\phi 1^2} + \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1})) LF_{3,2,-1}[m_2, \tilde{\mu}] - 4 C_{\phi 1^2} g_2^4 LF_{3,3,-3}[m_2, \tilde{\mu}] + \\
& 4 C_{\phi 1^2} g_2^4 m_2^2 LF_{3,3,-2}[m_2, \tilde{\mu}] - 4 C_{\phi 1^2} g_2^4 LF_{4,2,-3}[m_2, \tilde{\mu}] + 4 C_{\phi 1^2} g_2^4 m_2^2 LF_{4,2,-2}[m_2, \tilde{\mu}] - \\
& g_1^2 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} LF_{2,1,0}[m_d^r, m_q^p] - \frac{1}{2} \tilde{\mu} g_1^2 (\overline{C_{\phi 2\phi 1}} y_d^{pr} \overline{a_d^{pr}} + \overline{y_d^{pr}} (C_{\phi 2\phi 1} a_d^{pr} + 2 C_{\phi 1^2} \tilde{\mu} y_d^{pr})) \\
& LF_{2,2,0}[m_d^r, m_q^p] - g_1^2 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} LF_{2,1,0}[m_e^r, m_l^p] - \\
& \frac{1}{2} \tilde{\mu} g_1^2 (\overline{C_{\phi 2\phi 1}} y_e^{pr} \overline{a_e^{pr}} + \overline{y_e^{pr}} (C_{\phi 2\phi 1} a_e^{pr} + 2 C_{\phi 1^2} \tilde{\mu} y_e^{pr})) LF_{2,2,0}[m_e^r, m_l^p] + \\
& g_1^2 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} LF_{2,1,0}[m_l^p, m_e^r] - \frac{1}{2} \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} (g_1^2 + g_2^2) LF_{3,1,-1}[m_l^p, m_e^r] + \\
& \tilde{\mu} g_1^2 (\overline{C_{\phi 2\phi 1}} y_e^{pr} \overline{a_e^{pr}} + \overline{y_e^{pr}} (C_{\phi 2\phi 1} a_e^{pr} + 2 C_{\phi 1^2} \tilde{\mu} y_e^{pr})) LF_{3,1,0}[m_l^p, m_e^r] + \\
& \frac{1}{2} \tilde{\mu} g_1^2 (\overline{C_{\phi 2\phi 1}} y_e^{pr} \overline{a_e^{pr}} + \overline{y_e^{pr}} (C_{\phi 2\phi 1} a_e^{pr} + 2 C_{\phi 1^2} \tilde{\mu} y_e^{pr})) LF_{3,2,-1}[m_l^p, m_e^r] - \\
& \frac{3}{4} \tilde{\mu} (3g_1^2 + g_2^2) (\overline{C_{\phi 2\phi 1}} y_e^{pr} \overline{a_e^{pr}} + \overline{y_e^{pr}} (C_{\phi 2\phi 1} a_e^{pr} + 2 C_{\phi 1^2} \tilde{\mu} y_e^{pr})) LF_{4,1,-1}[m_l^p, m_e^r] + \\
& \frac{1}{3} \tilde{\mu} ((3g_1^2 + 2g_2^2) (C_{\phi 2\phi 1} \overline{y_e^{pr}} a_e^{pr} + \overline{C_{\phi 2\phi 1}} y_e^{pr} \overline{a_e^{pr}}) + 2 C_{\phi 1^2} \tilde{\mu} \overline{y_e^{pr}} y_e^{pr} (3g_1^2 + g_2^2)) \\
& LF_{5,1,-2}[m_l^p, m_e^r] + g_1^2 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} LF_{2,1,0}[m_q^p, m_d^r] - \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} (g_1^2 + g_2^2) \\
& LF_{3,1,-1}[m_q^p, m_d^r] + \tilde{\mu} g_1^2 (\overline{C_{\phi 2\phi 1}} y_d^{pr} \overline{a_d^{pr}} + \overline{y_d^{pr}} (C_{\phi 2\phi 1} a_d^{pr} + 2 C_{\phi 1^2} \tilde{\mu} y_d^{pr})) LF_{3,1,0}[m_q^p, m_d^r] + \\
& \frac{1}{2} \tilde{\mu} g_1^2 (\overline{C_{\phi 2\phi 1}} y_d^{pr} \overline{a_d^{pr}} + \overline{y_d^{pr}} (C_{\phi 2\phi 1} a_d^{pr} + 2 C_{\phi 1^2} \tilde{\mu} y_d^{pr})) LF_{3,2,-1}[m_q^p, m_d^r] - \\
& \frac{3}{4} \tilde{\mu} (5g_1^2 + 3g_2^2) (\overline{C_{\phi 2\phi 1}} y_d^{pr} \overline{a_d^{pr}} + \overline{y_d^{pr}} (C_{\phi 2\phi 1} a_d^{pr} + 2 C_{\phi 1^2} \tilde{\mu} y_d^{pr})) LF_{4,1,-1}[m_q^p, m_d^r] + \\
& \tilde{\mu} ((3g_1^2 + 2g_2^2) (C_{\phi 2\phi 1} \overline{y_d^{pr}} a_d^{pr} + \overline{C_{\phi 2\phi 1}} y_d^{pr} \overline{a_d^{pr}}) + 2 C_{\phi 1^2} \tilde{\mu} \overline{y_d^{pr}} y_d^{pr} (3g_1^2 + g_2^2)) \\
& LF_{5,1,-2}[m_q^p, m_d^r] - 3 \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} LF_{1,1,0}[m_q^p, m_q^s] - \\
& 6 C_{\phi 1^2} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} LF_{2,1,0}[m_q^p, m_q^s] + 6 C_{\phi 1^2} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} LF_{3,1,-1}[m_q^p, m_q^s] - \\
& \frac{1}{2} \overline{a_u^{pr}} a_u^{pr} (g_1^2 - 3g_2^2) LF_{2,1,0}[m_q^p, m_u^r] + \\
& \frac{1}{4} (-2 C_{\phi 1^2} \overline{a_u^{pr}} a_u^{pr} (g_1^2 - 7g_2^2) - \tilde{\mu} (g_1^2 - 5g_2^2) (C_{\phi 2\phi 1} \overline{y_u^{pr}} a_u^{pr} + \overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}})) \\
& LF_{2,2,0}[m_q^p, m_u^r] + \frac{3}{2} g_2^2 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} + \overline{a_u^{pr}} (4 C_{\phi 1^2} a_u^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{pr})) \\
& LF_{3,1,0}[m_q^p, m_u^r] - g_2^2 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} + \overline{a_u^{pr}} (4 C_{\phi 1^2} a_u^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{pr})) LF_{3,2,-1}[m_q^p, m_u^r] - \\
& \frac{3}{2} g_2^2 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} + \overline{a_u^{pr}} (4 C_{\phi 1^2} a_u^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{pr})) LF_{4,1,-1}[m_q^p, m_u^r] + \\
& \frac{1}{2} \overline{a_u^{pr}} a_u^{pr} (7g_1^2 + 3g_2^2) LF_{2,1,0}[m_u^r, m_q^p] - \frac{3}{2} \overline{a_u^{pr}} a_u^{pr} (g_1^2 + g_2^2) LF_{3,1,-1}[m_u^r, m_q^p] + \\
& (C_{\phi 1^2} \overline{a_u^{pr}} a_u^{pr} (7g_1^2 + g_2^2) + \frac{1}{2} \tilde{\mu} (7g_1^2 + 2g_2^2) (C_{\phi 2\phi 1} \overline{y_u^{pr}} a_u^{pr} + \overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}})) LF_{3,1,0}[m_u^r, \\
& m_q^p] + \frac{1}{4} (2 C_{\phi 1^2} \overline{a_u^{pr}} a_u^{pr} (g_1^2 - 7g_2^2) + \tilde{\mu} (g_1^2 - 5g_2^2) (C_{\phi 2\phi 1} \overline{y_u^{pr}} a_u^{pr} + \overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}})) \\
& LF_{3,2,-1}[m_u^r, m_q^p] + (-3 C_{\phi 1^2} \overline{a_u^{pr}} a_u^{pr} (5g_1^2 + g_2^2) - \\
& \frac{3}{2} \tilde{\mu} (5g_1^2 + 2g_2^2) (C_{\phi 2\phi 1} \overline{y_u^{pr}} a_u^{pr} + \overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}})) LF_{4,1,-1}[m_u^r, m_q^p] + \\
& (2 C_{\phi 1^2} \overline{a_u^{pr}} a_u^{pr} (3g_1^2 + g_2^2) + \tilde{\mu} (3g_1^2 + 2g_2^2) (C_{\phi 2\phi 1} \overline{y_u^{pr}} a_u^{pr} + \overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}})) \\
& LF_{5,1,-2}[m_u^r, m_q^p] - 3 \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} LF_{1,1,0}[m_u^r, m_u^t] - \\
& \frac{3}{4} g_1^2 (g_1^2 + g_2^2) LF_{2,1,-1}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^2 (g_1^2 + g_2^2) LF_{3,1,-2}[\tilde{\mu}, m_1] - \\
& 2 C_{\phi 1^2} g_1^2 (g_1^2 + g_2^2) LF_{3,1,-1}[\tilde{\mu}, m_1] + \frac{1}{2} m_1 \tilde{\mu} g_1^2 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (g_1^2 + g_2^2) LF_{3,1,0}[\tilde{\mu}, m_1] + \\
& \frac{3}{2} C_{\phi 1^2} g_1^4 LF_{3,2,-2}[\tilde{\mu}, m_1] + \frac{1}{2} m_1 g_1^4 (3 m_1 C_{\phi 1^2} - 2 \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1})) LF_{3,2,-1}[\tilde{\mu}, m_1] + \\
& \frac{1}{3} C_{\phi 1^2} g_1^2 (12 g_1^2 + 7 g_2^2) LF_{4,1,-2}[\tilde{\mu}, m_1] - \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (6 g_1^2 + 5 g_2^2) LF_{4,1,-1}[\tilde{\mu}, m_1] - C_{\phi 1^2} g_1^4 LF_{4,2,-3}[\tilde{\mu}, m_1] + \\
& m_1 g_1^4 (-m_1 C_{\phi 1^2} + \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1})) LF_{4,2,-2}[\tilde{\mu}, m_1] - \frac{2}{3} C_{\phi 1^2} g_1^2 (3 g_1^2 + g_2^2) LF_{5,1,-3}[\tilde{\mu}, m_1] + \\
& \frac{1}{3} m_1 \tilde{\mu} g_1^2 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (3 g_1^2 + 2 g_2^2) LF_{5,1,-2}[\tilde{\mu}, m_1] - \frac{9}{4} g_2^2 (g_1^2 + g_2^2) LF_{2,1,-1}[\tilde{\mu}, m_2] + \\
& \frac{3}{2} g_2^2 (g_1^2 + g_2^2) LF_{3,1,-2}[\tilde{\mu}, m_2] - \frac{2}{3} C_{\phi 1^2} g_2^2 (9 g_1^2 + g_2^2) LF_{3,1,-1}[\tilde{\mu}, m_2] + \\
& \frac{1}{6} m_2 \tilde{\mu} g_2^2 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (9 g_1^2 + 5 g_2^2) LF_{3,1,0}[\tilde{\mu}, m_2] + \frac{85}{6} C_{\phi 1^2} g_2^4 LF_{3,2,-2}[\tilde{\mu}, m_2] + \\
& \frac{1}{6} m_2 g_2^4 (9 m_2 C_{\phi 1^2} - 28 \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1})) LF_{3,2,-1}[\tilde{\mu}, m_2] + \\
& 3 C_{\phi 1^2} g_2^2 (4 g_1^2 + g_2^2) LF_{4,1,-2}[\tilde{\mu}, m_2] - \\
& \frac{1}{4} m_2 \tilde{\mu} g_2^2 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (18 g_1^2 + 11 g_2^2) LF_{4,1,-1}[\tilde{\mu}, m_2] - 7 C_{\phi 1^2} g_2^4 LF_{4,2,-3}[\tilde{\mu}, m_2] + \\
& m_2 g_2^4 (-m_2 C_{\phi 1^2} + 4 \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1})) LF_{4,2,-2}[\tilde{\mu}, m_2] - 2 C_{\phi 1^2} g_2^2 (3 g_1^2 + g_2^2) LF_{5,1,-3}[\tilde{\mu}, m_2] + \\
& m_2 \tilde{\mu} g_2^2 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (3 g_1^2 + 2 g_2^2) LF_{5,1,-2}[\tilde{\mu}, m_2] + 8 C_{\phi 1^2} g_1^2 g_2^2 LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\
& \frac{1}{2} g_1^2 g_2^2 (12 m_1 m_2 C_{\phi 1^2} - \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (7 m_1 + 5 m_2)) LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
& 4 C_{\phi 1^2} g_1^2 g_2^2 LF_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] + \\
& 2 g_1^2 g_2^2 (-2 m_1 m_2 C_{\phi 1^2} + \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (m_1 + m_2)) LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\
& 6 C_{\phi 1^2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} LF_{2,2,1,-1}[m_q^p, m_q^s, m_d^r] - 6 \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} a_u^{sr} \\
& LF_{1,1,1,0}[m_q^p, m_q^s, m_u^r] - \frac{3}{2} \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} (4 C_{\phi 1^2} a_u^{sr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr}) LF_{2,1,1,0}[m_q^p, m_q^s, m_u^r] + \\
& \frac{3}{2} \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} (4 C_{\phi 1^2} a_u^{sr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr}) LF_{3,1,1,-1}[m_q^p, m_q^s, m_u^r] - \\
& \frac{3}{2} C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{sr} a_u^{pt} LF_{2,1,1,0}[m_q^p, m_q^s, m_u^t] + \\
& \frac{3}{2} C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{sr} a_u^{pt} LF_{3,1,1,-1}[m_q^p, m_q^s, m_u^t] + \\
& \frac{3}{4} \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} (4 C_{\phi 1^2} a_u^{sr} + 3 \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr}) LF_{2,2,1,-1}[m_q^p, m_u^r, m_q^s] - \\
& 6 \overline{y_u^{st}} y_u^{sr} \overline{a_u^{pr}} a_u^{pt} LF_{1,1,1,0}[m_q^p, m_u^r, m_u^t] + \\
& \frac{9}{4} C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{sr} a_u^{pt} LF_{2,2,1,-1}[m_q^p, m_u^t, m_q^s] + \\
& \frac{3}{2} \tilde{\mu} (C_{\phi 2\phi 1} \overline{y_d^{pr}} \overline{y_u^{st}} y_u^{pt} a_d^{sr} + \overline{C_{\phi 2\phi 1}} y_d^{pr} \overline{y_u^{pt}} y_u^{st} \overline{a_d^{sr}}) LF_{2,2,1,-1}[m_q^s, m_q^p, m_d^r] - \\
& \frac{3}{2} \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} (4 C_{\phi 1^2} a_u^{sr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr}) LF_{2,1,1,0}[m_q^s, m_q^p, m_u^r] + \\
& \frac{3}{2} \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} (4 C_{\phi 1^2} a_u^{sr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr}) LF_{3,1,1,-1}[m_q^s, m_q^p, m_u^r] - \\
& \frac{3}{2} C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{sr} a_u^{pt} LF_{2,1,1,0}[m_q^s, m_q^p, m_u^t] + \\
& \frac{3}{2} C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{sr} a_u^{pt} LF_{3,1,1,-1}[m_q^s, m_q^p, m_u^t] + \\
& \frac{3}{4} \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} (4 C_{\phi 1^2} a_u^{sr} - \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr}) LF_{2,2,1,-1}[m_q^s, m_u^r, m_q^p] - \\
& \frac{3}{4} C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{sr} a_u^{pt} LF_{2,2,1,-1}[m_q^s, m_u^t, m_q^p] - \\
& 3 \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} (2 C_{\phi 1^2} a_u^{sr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr}) LF_{2,1,1,0}[m_u^r, m_q^p, m_q^s] + \\
& 3 \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} (2 C_{\phi 1^2} a_u^{sr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr}) LF_{3,1,1,-1}[m_u^r, m_q^p, m_q^s] - \\
& 6 \overline{y_u^{st}} y_u^{sr} a_u^{pt} (C_{\phi 1^2} \overline{a_u^{pr}} + C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}}) LF_{2,1,1,0}[m_u^r, m_q^p, m_u^t] + \\
& 6 \overline{y_u^{st}} y_u^{sr} a_u^{pt} (C_{\phi 1^2} \overline{a_u^{pr}} + C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}}) LF_{3,1,1,-1}[m_u^r, m_q^p, m_u^t] + \\
& 3 \overline{y_u^{st}} y_u^{sr} (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} a_u^{pt} + \overline{a_u^{pr}} (2 C_{\phi 1^2} a_u^{pt} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{pt})) LF_{2,2,1,-1}[m_u^r, m_u^t, m_q^p] - \\
& 3 C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{sr} a_u^{pt} LF_{2,1,1,0}[m_u^t, m_q^p, m_q^s] + \\
& 3 C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{sr} a_u^{pt} LF_{3,1,1,-1}[m_u^t, m_q^p, m_q^s] - \\
& 6 \overline{y_u^{st}} y_u^{sr} \overline{a_u^{pr}} (C_{\phi 1^2} a_u^{pt} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{pt}) LF_{2,1,1,0}[m_u^t, m_q^p, m_u^r] + \\
& 6 \overline{y_u^{st}} y_u^{sr} \overline{a_u^{pr}} (C_{\phi 1^2} a_u^{pt} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{pt}) LF_{3,1,1,-1}[m_u^t, m_q^p, m_u^r] + \\
& g_1^2 g_2^2 LF_{2,1,1,-2}[\tilde{\mu}, m_1, m_2] + g_1^2 g_2^2 (-3 C_{\phi 1^2} + m_1 m_2) LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_2] + \\
& m_1 \tilde{\mu} g_1^2 g_2^2 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{2,1,1,0}[\tilde{\mu}, m_1, m_2] + 7 C_{\phi 1^2} g_1^2 g_2^2 LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_2] + \\
& \frac{1}{4} g_1^2 g_2^2 (12 m_1 m_2 C_{\phi 1^2} - \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (9 m_1 + 5 m_2)) LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] - \\
& 4 C_{\phi 1^2} g_1^2 g_2^2 LF_{4,1,1,-3}[\tilde{\mu}, m_1, m_2] + \frac{1}{2} g_1^2 g_2^2 (-4 m_1 m_2 C_{\phi 1^2} + 3 \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (m_1 + m_2)) \\
& LF_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - 4 C_{\phi 1^2} g_1^2 g_2^2 LF_{3,2,1,-3}[\tilde{\mu}, m_2, m_1] + \\
& 2 g_1^2 g_2^2 (-2 m_1 m_2 C_{\phi 1^2} + \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) (m_1 + m_2)) LF_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] - \\
& 3 \tilde{\mu}^4 \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{pt} y_d^{sr} LF_{1,1,1,1,0}[m_d^r, m_d^t, m_q^p, m_q^s] + \\
& \frac{3}{2} \tilde{\mu}^3 \overline{y_d^{st}} y_d^{sr} (C_{\phi 2\phi 1} \overline{y_d^{pr}} a_d^{pt} - \overline{C_{\phi 2\phi 1}} y_d^{pt} \overline{a_d^{pr}}) LF_{2,2,1,1,-1}[m_d^r, m_q^p, m_d^t, m_q^s] + \\
& \frac{3}{2} \tilde{\mu}^3 \overline{y_d^{st}} y_d^{sr} (-C_{\phi 2\phi 1} \overline{y_d^{pr}} a_d^{pt} + \overline{C_{\phi 2\phi 1}} y_d^{pt} \overline{a_d^{pr}}) LF_{2,2,1,1,-1}[m_d^t, m_q^p, m_d^r, m_q^s] - \\
& \tilde{\mu}^4 \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{pt} y_e^{sr} LF_{1,1,1,1,0}[m_e^r, m_e^t, m_l^p, m_l^s] + \\
& \frac{1}{2} \tilde{\mu}^3 \overline{y_e^{st}} y_e^{sr} (C_{\phi 2\phi 1} \overline{y_e^{pr}} a_e^{pt} - \overline{C_{\phi 2\phi 1}} y_e^{pt} \overline{a_e^{pr}}) LF_{2,2,1,1,-1}[m_e^r, m_l^p, m_e^t, m_l^s] + \\
& \frac{1}{2} \tilde{\mu}^3 \overline{y_e^{st}} y_e^{sr} (-C_{\phi 2\phi 1} \overline{y_e^{pr}} a_e^{pt} + \overline{C_{\phi 2\phi 1}} y_e^{pt} \overline{a_e^{pr}}) LF_{2,2,1,1,-1}[m_e^t, m_l^p, m_e^r, m_l^s] - \\
& \tilde{\mu}^3 \overline{y_e^{st}} y_e^{sr} (\overline{C_{\phi 2\phi 1}} y_e^{pt} \overline{a_e^{pr}} + \overline{y_e^{pr}} (C_{\phi 2\phi 1} a_e^{pt} + 2 C_{\phi 1^2} \tilde{\mu} y_e^{pt})) LF_{2,1,1,1,0}[m_l^p, m_e^r, m_e^t, m_l^s] + \\
& \tilde{\mu}^3 \overline{y_e^{st}} y_e^{sr} (\overline{C_{\phi 2\phi 1}} y_e^{pt} \overline{a_e^{pr}} + \overline{y_e^{pr}} (C_{\phi 2\phi 1} a_e^{pt} + 2 C_{\phi 1^2} \tilde{\mu} y_e^{pt})) LF_{3,1,1,1,-1}[m_l^p, m_e^r, m_e^t, m_l^s] - \\
& 3 \tilde{\mu}^3 \overline{y_d^{st}} y_d^{sr} (\overline{C_{\phi 2\phi 1}} y_d^{pt} \overline{a_d^{pr}} + \overline{y_d^{pr}} (C_{\phi 2\phi 1} a_d^{pt} + 2 C_{\phi 1^2} \tilde{\mu} y_d^{pt})) LF_{2,1,1,1,0}[m_q^p, m_d^r, m_d^t, m_q^s] + \\
& 3 \tilde{\mu}^3 \overline{y_d^{st}} y_d^{sr} (\overline{C_{\phi 2\phi 1}} y_d^{pt} \overline{a_d^{pr}} + \overline{y_d^{pr}} (C_{\phi 2\phi 1} a_d^{pt} + 2 C_{\phi 1^2} \tilde{\mu} y_d^{pt})) LF_{3,1,1,1,-1}[m_q^p, m_d^r, m_d^t, \\
& m_q^s] + \frac{3}{2} \tilde{\mu} y_d^{sr} a_u^{pt} (\overline{C_{\phi 2\phi 1}} \overline{a_d^{pr}} \overline{a_u^{st}} + C_{\phi 2\phi 1} \tilde{\mu}^2 \overline{y_d^{pr}} \overline{y_u^{st}}) LF_{2,2,1,1,-1}[m_q^p, m_q^s, m_d^r, m_u^t] + \\
& \frac{3}{2} \tilde{\mu} \overline{y_d^{st}} \overline{a_u^{pr}} (C_{\phi 2\phi 1} a_d^{pt} a_u^{sr} + \tilde{\mu} y_d^{pt} (4 C_{\phi 1^2} a_u^{sr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr})) \\
& LF_{2,2,1,1,-1}[m_q^p, m_q^s, m_d^t, m_u^r] - 3 \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} a_u^{sr} LF_{1,1,1,1,0}[m_q^p, m_q^s, m_u^r, m_u^t] - \\
& \frac{3}{2} (\overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{st} \overline{a_u^{pt}} \overline{a_u^{sr}} a_u^{pr} + \overline{a_u^{pr}} a_u^{pt} a_u^{sr} (4 C_{\phi 1^2} \overline{a_u^{st}} + C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{st}})) \\
& LF_{2,1,1,1,0}[m_q^p, m_q^s, m_u^r, m_u^t] + 6 C_{\phi 1^2} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} a_u^{sr} LF_{3,1,1,1,-1}[m_q^p, m_q^s, m_u^r, m_u^t] + \\
& \frac{3}{2} \tilde{\mu} (C_{\phi 2\phi 1} \overline{y_u^{st}} \overline{a_u^{pr}} a_u^{pt} a_u^{sr} + \overline{C_{\phi 2\phi 1}} y_u^{st} \overline{a_u^{pt}} \overline{a_u^{sr}} a_u^{pr}) LF_{3,1,1,1,-1}[m_q^p, m_q^s, m_u^t, m_u^r] + \\
& \frac{3}{2} \tilde{\mu} (C_{\phi 2\phi 1} \overline{y_u^{st}} \overline{a_u^{pr}} a_u^{pt} a_u^{sr} + \overline{C_{\phi 2\phi 1}} y_u^{st} \overline{a_u^{pt}} \overline{a_u^{sr}} a_u^{pr}) LF_{3,1,1,1,-1}[m_q^s, m_q^p, m_u^t, m_u^r] + \\
& \frac{3}{4} \tilde{\mu} \overline{a_u^{pr}} a_u^{pt} (C_{\phi 2\phi 1} \overline{y_u^{st}} a_u^{sr} - \overline{C_{\phi 2\phi 1}} y_u^{sr} \overline{a_u^{st}}) LF_{2,2,1,1,-1}[m_q^s, m_u^r, m_q^p, m_u^t] + \\
& \frac{3}{4} \tilde{\mu} \overline{a_u^{pr}} a_u^{pt} (-C_{\phi 2\phi 1} \overline{y_u^{st}} a_u^{sr} + \overline{C_{\phi 2\phi 1}} y_u^{sr} \overline{a_u^{st}}) LF_{2,2,1,1,-1}[m_q^s, m_u^t, m_q^p, m_u^r] - \\
& 3 \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} (2 C_{\phi 1^2} a_u^{sr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr}) LF_{2,1,1,1,0}[m_u^r, m_q^p, m_q^s, m_u^t] + \\
& 3 \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} (2 C_{\phi 1^2} a_u^{sr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr}) LF_{3,1,1,1,-1}[m_u^r, m_q^p, m_q^s, m_u^t] + \\
& \frac{3}{2} \overline{a_u^{pr}} a_u^{pt} (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{st}} a_u^{sr} + \overline{a_u^{st}} (2 C_{\phi 1^2} a_u^{sr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_u^{sr})) \\
& LF_{2,2,1,1,-1}[m_u^r, m_u^t, m_q^p, m_q^s] - 3 C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{st}} \overline{a_u^{pr}} a_u^{pt} a_u^{sr} LF_{2,1,1,1,0}[m_u^t, m_q^p, m_q^s, m_u^r] + \\
& 3 C_{\phi 2\phi 1} \tilde{\mu} \overline{y_u^{st}} \overline{a_u^{pr}} a_u^{pt} a_u^{sr} LF_{3,1,1,1,-1}[m_u^t, m_q^p, m_q^s, m_u^r])
\end{aligned}$$